



Functional Description

Gas engine-generator sets 12V400 GS
with engines **E3042Z6** and **B3042Z7**

MS13067/00E

Name	SLU Name Model	Engine model	Application group
MTU 12V400 GS	GG12V0400C1	E3042Z6	3A, continuous operation, unrestricted
MTU 12V400 GS	GG12V0400C1	B3042Z7	3A, continuous operation, unrestricted

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Table of Contents

1 Engine-Generator Set		8.2 Cooling system with heat supply from cogeneration from engine coolant	40
1.1 Application group	4	8.3 Cooling system with heat supply from cogeneration from engine coolant and exhaust gas	41
1.2 Nameplate and designation	5	8.4 Heat module	42
1.3 Scope of delivery - Genset without acoustic enclosure	8	8.5 Heating water system	46
1.4 Scope of delivery - Genset with acoustic enclosure	12	9 Electrical System	
2 Engine		9.1 Generator power control panel (GLF)	48
2.1 Engine - Overview	15	9.2 Starter battery	49
2.2 Engine side and cylinder designations	18	9.3 Battery charger	51
2.3 Firing order	19	10 Gas alarm and fire alarm	
3 Generator		10.1 Gas detector	52
3.1 Generator	20	10.2 Fire detection system	53
4 Monitoring and Control		11 Technical Data	
4.1 MMC and MIP control system	22	11.1 Technical data	54
5 Charge-air system and gas system		12 Appendix A	
5.1 Mixture formation and supercharging	24	12.1 Abbreviations	55
5.2 Gas train	28	13 Index	
6 Exhaust System		13.1 Index	58
6.1 Exhaust system	30		
6.2 Exhaust gas heat exchanger	33		
7 Lube Oil System			
7.1 Lube oil system	34		
8 Cooling system and hot water system			
8.1 Cooling system without heat supply from cogeneration	38		

1 Engine-Generator Set

1.1 Application group

The genset consists of a gas engine and a generator mounted on a common baseframe. Upon demand, the engine will start and drives the generator to generate heat energy and to produce electrical power. On the version without a heat module, no heat energy is generated.

Application group 3A – Continuous operation, unrestricted (Continuous Power)

Gensets for unrestricted continuous power and heat energy supply are used for:

- Base load generation
- Part of a power supply or the public grid

Continuous operation	Application group 3A
Operating mode	Continuous operation, unrestricted
Rating definition	10% overload capacity (ICXN)
Load factor	< 100 %
Operating hours	Unrestricted

Benefits

- Wide range of standardized gensets to meet customer requirements with regard to power, emission and other characteristics.
- Option of selecting different components:
 - Only current supply
 - Heat supply from cogeneration from engine coolant
 - Heat supply from cogeneration from engine coolant and exhaust gas
- Most recent gas engine technology
- Main components with state-of-the-art technology for high electrical and thermal efficiency and long service life.

1.2 Nameplate and designation

Nameplates on genset

The genset has the following nameplates:

- Nameplate of genset
- Engine nameplate
- Nameplate of generator

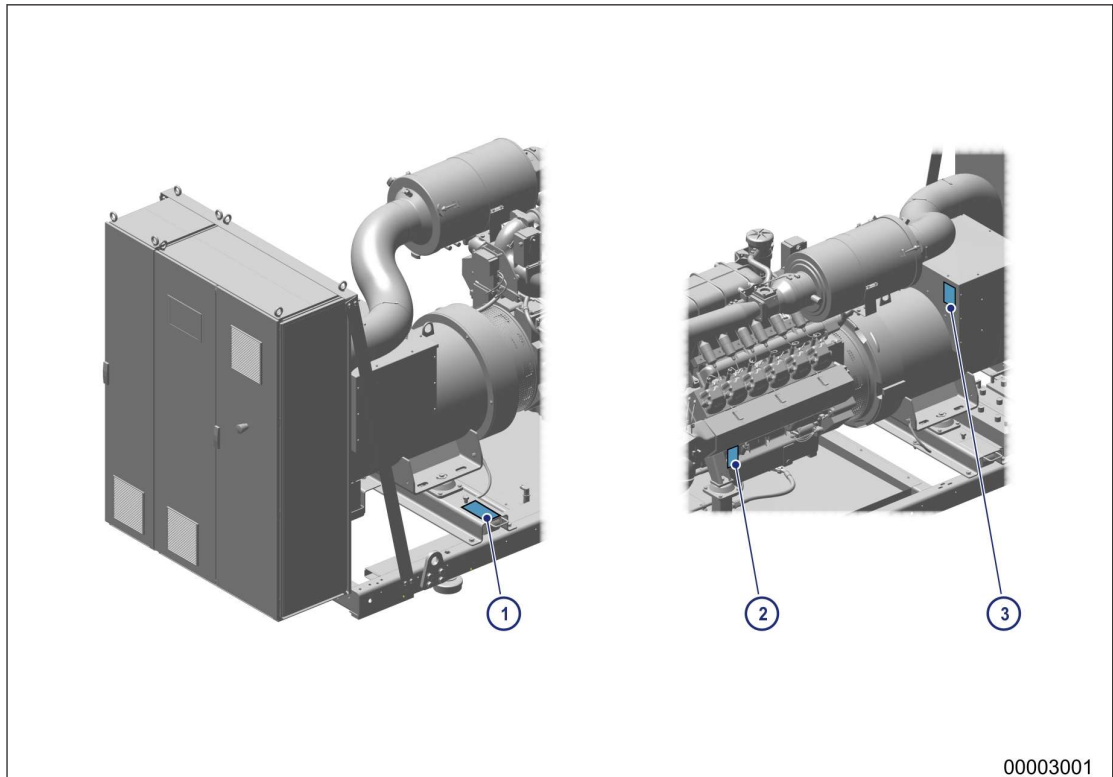


Figure 1: Fitting position of nameplates

1 Nameplate of genset

2 Engine nameplate

3 Nameplate of generator

Explanation of product name

Example: MTU 12V400 GS

MTU	Manufacturer: MTU Onsite Energy
12	Number of cylinders of the engine
V	Cylinder arrangement of the engine: Vee arrangement
400	MTU Series 400
GS	Gas System

Overview of application groups on nameplate

Specification on the nameplate	Application group
CONTINUOUS POWER	3A – Continuous operation, unrestricted

Genset – nameplate and designation

**Stromerzeugungsaggregat
Genset**

Leistungsangaben nach ISO-Standard Leistung ISO 3046-1 nicht überlastbar
 Luftdruck (Gesamtdruck) $p_r = 100 \text{ kPa}$ (100 ü.N.N.)
 Lufttemperatur (am Luftfiltereintritt) $T_r = 298 \text{ K}$ ($t_r = 25^\circ \text{C}$)
 Relative Luftfeuchte $\phi = 30\%$
 Standard reference conditions ISO standard power ISO 3046-1 no overload
 Total barometric pressure $p_r = 100 \text{ kPa}$ (100 m a. s. l.)
 Air temperature $T_r = 298 \text{ K}$ ($t_r = 25^\circ \text{C}$)
 Relative humidity $\phi = 30\%$

Typ
 Type

Modul- / Aggregat- Nr.
 Module / Genset No.

Projekt Nr.
 Project No.

Baujahr
 Year of Manufacture

Gewicht mit / ohne Betriebsstoffe [kg]
 Weight with / without fluids and lubricants [kg]

Elektrische Leistung COP [kW]
 Electrical Power COP [kW]

Performance- Class

Nennspannung [V]
 Nominal Voltage [V]

Nennfrequenz [Hz]
 Nominal Frequency [Hz]

Max. Außentemperatur [°C]
 Max. outdoor temperature [°C]

Nennleistungsfaktor $\cos \phi$ induktiv / kapazitiv
 Nominal Rated Power Factor $\cos \phi$ inductive / capacitive

Max. Aufstellhöhe [m ü.N.N.]
 Max. installation height [m a.s.l.]

**Wärmeauskopplung nach DIN 6280
 Heat extraction to DIN 6280**

Thermische Nennleistung [kW]
 Thermal output [kW]

Zul. Heizwasser-temperatur [°C]
 Admissible heating water temperature [°C] min./max.

Zul. Betriebsdruck im Heizwasserkreis
 Admissible operating pressure in heating water circuit

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 Dasinger Str. 11
 D-86165 Augsburg

CE
 Made in Germany

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Figure 2: Nameplate of genset

Item	Designation
1	Power ratings as per ISO standard power ISO 3046-1 no overload capability Air pressure (total pressure) $p_r = 100 \text{ kPa}$ (100 above sea level) Air temperature (at air filter inlet) $T_r = 298 \text{ K}$ ($t_r = 25^\circ \text{C}$) Relative air humidity $\phi = 30\%$
2	Heat supply from cogeneration as per DIN 6280
3	MTU Onsite Energy Dasinger Str. 11 D-86165 Augsburg
4	Type
5	Module / genset no.
6	Project no.
7	Year of construction
8	Weight with / without fluids and lubricants [kg]
9	Electrical output COP [KW]
10	Performance class
11	Rated voltage [V]
12	Rated frequency [Hz]
13	Max. outside temperature [°C]
14	Rated power factor $\cos \phi$ inductive / capacitive
15	Max. site altitude [m above sea level]
16	Thermal rated power [kW]

Item	Designation
17	Perm. heating water temperature [°C]
18	Perm. operating pressure in heating water circuit

1.3 Scope of delivery – Genset without acoustic enclosure

Genset without acoustic enclosure

The genset comprises the following main components:

- Engine
- Gas train
- Generator and coupling
- Baseframe
- Switch cabinet with:
 - MMC (MTU Module Control) for controlling the genset
 - MIP (MTU Interface Panel) for engine control

In the combustion engine, the energy contained in the fuel is converted into mechanical work and heat.

The generator, which converts the mechanical work into electrical energy, is flange-mounted via a resilient coupling on the engine. The engine and generator are joined via resilient, vibration-damped elements to the baseframe.

All closed-loop control, open-loop control and monitoring systems, including communication options from and to the outside, are implemented by means of the hardware and software in the switch cabinet.

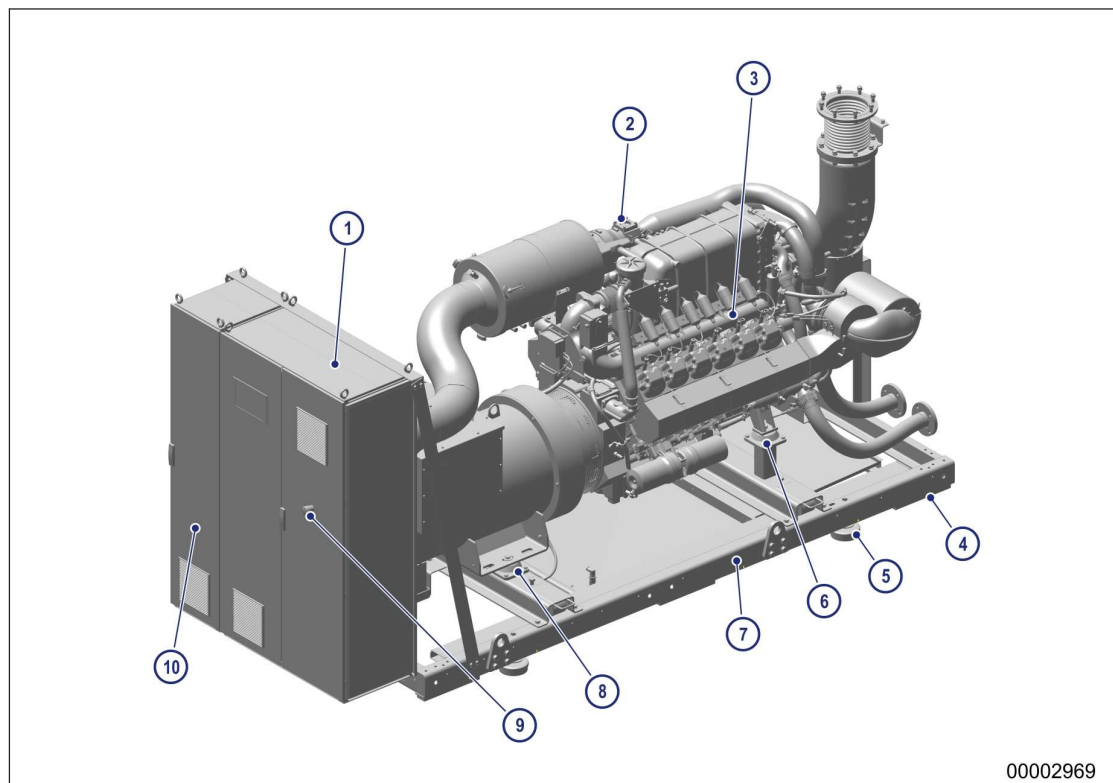
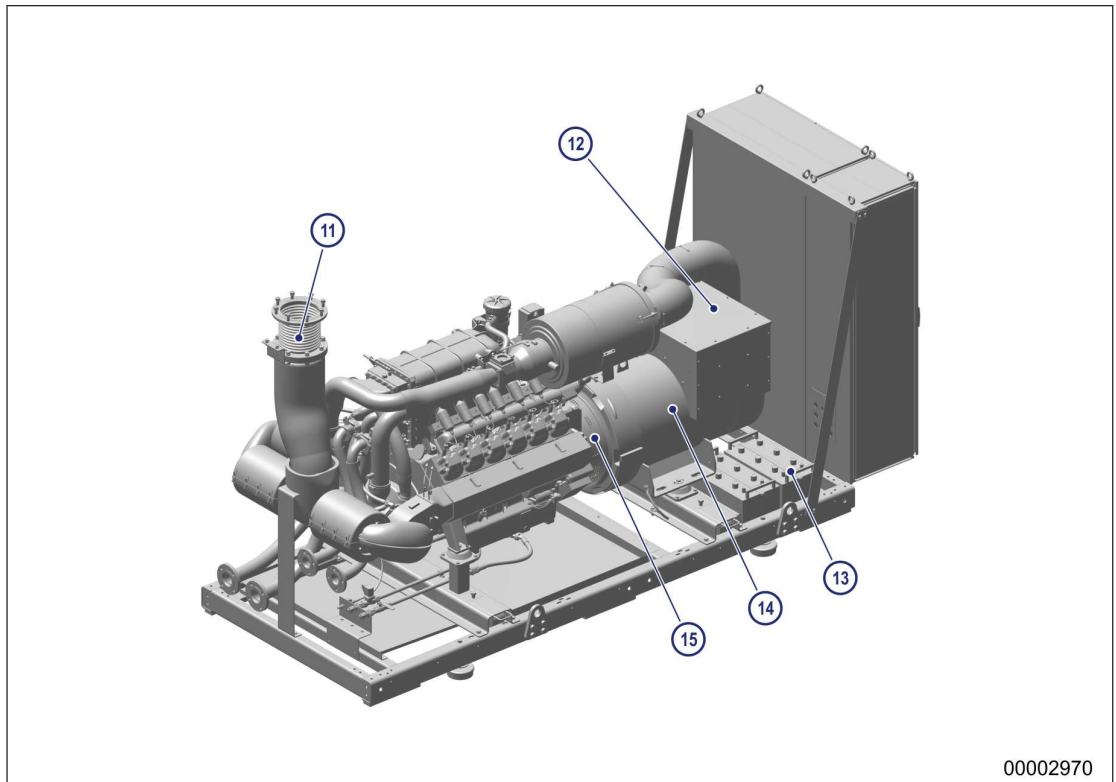


Figure 3: Genset – generator end

- | | | |
|---|---------------------------|---|
| 1 Switch cabinet for MMC control system | 5 Machine shoe (optional) | 9 EMERGENCY-STOP button |
| 2 Gas mixer (gas train connection) | 6 Engine mounting | 10 Generator power control panel (GLF) (optional) |
| 3 Engine | 7 Baseframe | |
| 4 Baseframe mount (optional) | 8 Generator mounting | |

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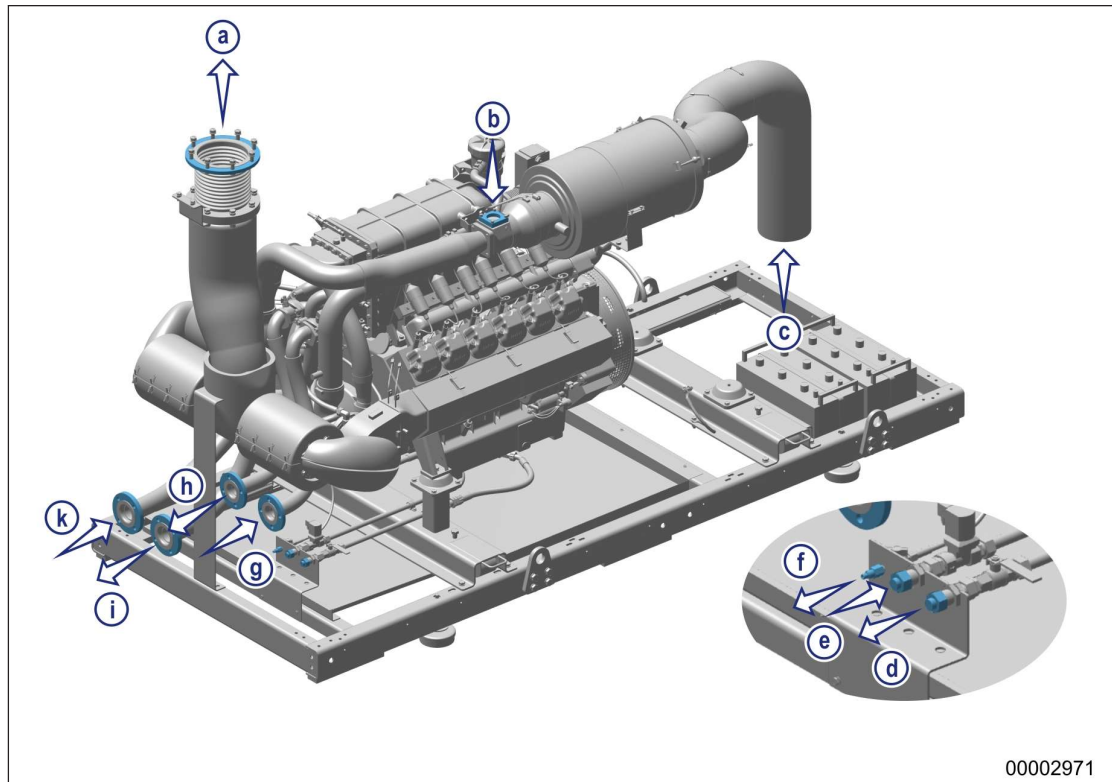
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Figure 4: Genset – engine end

- 11 Exhaust pipe bellows (optional)
- 12 Generator terminal box

- 13 Starter battery (optional)
- 14 Generator

- 15 Coupling



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Figure 5: Genset – connections

- | | | |
|--------------------|---|-------------------------|
| a Exhaust outlet | e Lube oil inlet | i Engine coolant outlet |
| b Gas inlet | f Lube oil outlet via oil spraying system | k Engine coolant inlet |
| c Intake air inlet | g Mixture coolant inlet | |
| d Lube oil outlet | h Mixture coolant outlet | |

The connection hoses for g, h, i and k can be included in the scope of delivery as options.

- Gas engine of the MTU 400 Series:
 - Rated speed 1,500 rpm (50 Hz)
 - Electric starters
- Generator:
 - Voltage 400 V, 50 Hz
 - 2 bearings
 - Digital voltage regulator
 - Measurement current transformer x/1A
 - Protection current transformer x/1A 5P10
 - Winding temperature monitor 2 PT100 per winding
 - Bearing temperature monitor 1 PT100 per bearing
 - Anti-condensation heating
 - Degree of protection IP23
- Gas train for natural gas or biogas
- Lube oil supply system:
 - Oil spraying system for extending the oil service life
 - Control of oil system with MMC control system
- MMC control system (MTU Modul Control):
 - MIP engine control (MTU Interface Panel)
 - Connection to emergency-stop button
- Air filter
- Coupling
- Safety guards for all rotating parts
- Baseframe with resilient engine and generator mounts

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